

British Airways AI System Architecture

Comprehensive Design Guide: The Heathrow Airport Model, 5 Specialized Layers & Triage Engine

TARGET CARRIER

British Airways (@British_Airways)

SYSTEM LAYERS

5 Decoupled Engineering Layers

VECTOR STORE

ChromaDB (23,859 Paired Cases)

TRIAGE STRATEGY

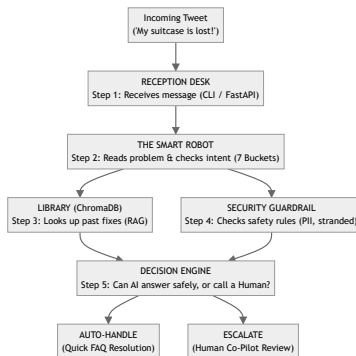
Hybrid Guardrails + LLM Reasoner

British Airways AI Customer Support & Triage Agent

A comprehensive, visual, and beginner-friendly guide to how our AI system works under the hood.

1. The Big Picture: The Airport Helpdesk Analogy

Imagine you are standing in a bustling terminal at **London Heathrow Airport (LHR)**. Thousands of passengers arrive every hour with questions, complaints, lost bags, and flight delays.

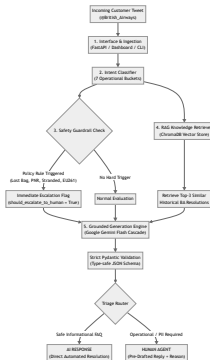


In an airline helpdesk, you have two types of requests: 1. **Simple, Safe Questions**: "How heavy can my carry-on bag be?" The smart robot knows the answer by heart. It replies in 3 seconds. The human staff never have to lift a finger. 2. **High-Stakes Crises**: "My flight was cancelled, I'm stuck in London with no hotel, and here is my booking reference KL92X1!" The robot never tries to handle this alone. It catches the sensitive booking code, flags the emergency, writes a helpful draft apology, and hands the ticket directly to a human airline supervisor.

This is called a **Hybrid Triage & Resolution Architecture**.

2. End-to-End System Workflow

Here is how a message travels through our 5-layer pipeline from the second a customer tweets to the moment a resolution is drafted:



Detailed Triage Pathway Specifications: * **Autonomous AI Resolution Pathway**: [AI_Handle.md](#) * **Human-in-the-Loop Co-Pilot Pathway**: [Human_Agent.md](#)

3. The 5 Architectural Layers Explained

Layer 1: The Reception Desk (Interface Layer)

- Detailed Specification**: [Reception_Desk.md](#)
- What it does**: Welcomes and validates incoming customer queries.
- Components**:
 - Interactive Web Dashboard ([dashboard.html](#))**: A production-grade visual Copilot inbox at [/dashboard](#) with scenario filtering, RAG precedent inspection, and architecture views.
 - Interactive CLI ([run_demo.py](#))**: A colorful terminal interface for instant manual testing of custom and preset tweets.
 - FastAPI Microservice ([src/api.py](#))**: A production-grade REST API with interactive Swagger documentation ([/docs](#)) exposing the [POST /api/v1/triage](#) endpoint.
- Why it matters**: Whether it's a web dashboard, batch script, or REST service, all call the exact same underlying agent engine without code duplication.

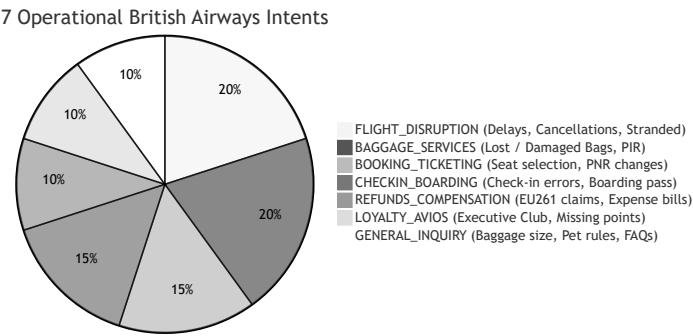
Layer 2: The Memory Vault (Knowledge Retrieval / RAG Layer)

- **Detailed Specification:** [Memory Vault.md](#)
- **What it does:** Gives the AI an institutional memory of how real British Airways agents have solved problems in the past.
- **How it works:**
 - We took **23,859 historical conversations** between real customers and `@British_Airways` from Twitter.
 - We converted 1,500 representative customer questions into mathematical coordinates (called **vector embeddings**) using `gemini-embedding-2` and `all-MiniLM-L6-v2`.
 - We stored them in a local, embedded database called **ChromaDB**.
 - When a new customer asks: "Where is my suitcase?", ChromaDB searches for past tweets asking about missing luggage and retrieves the exact solutions British Airways agents used!



Layer 3: The 7-Intent Classifier

- **Detailed Specification:** [Intent Classification.md](#) Instead of confusing the model with dozens of overlapping categories, we group all airline customer requests into **7 crystal-clear operational buckets**:

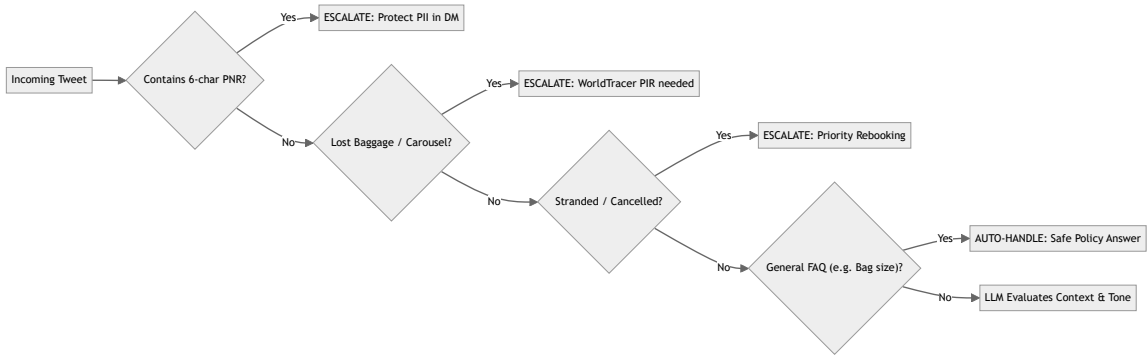


Intent Name	Example Customer Tweet	What It Means
FLIGHT_DISRUPTION	"My flight BA178 was cancelled and I am stranded!"	Passenger is travelling right now and needs urgent rebooking.
BAGGAGE_SERVICES	"My suitcase didn't come out on carousel 4."	Baggage desk needs to trace the bag via WorldTracer.
BOOKING_TICKETING	"Can you change my seat on booking ref KL92X1?"	Itinerary change requiring 6-character PNR access.
CHECKIN_BOARDING	"The BA app won't open my boarding pass at gate B32."	Airport check-in or boarding barcode failure.
REFUNDS_COMPENSATION	"I was delayed 5 hours, pay my statutory EU261 claim!"	Legal financial claim or cash reimbursement.
LOYALTY_AVIOS	"My Executive Club account is missing 80 tier points."	Frequent flyer account investigation.
GENERAL_INQUIRY	"What is the cabin bag allowance for Euro Traveller?"	Routine airline policy question (Safe to Auto-Handle).

Layer 4: The Hybrid Triage Engine (Safety Guardrails)

- **Detailed Specification:** [Hybrid Triage Guardrails.md](#)

Why do we call it "Hybrid"?
Because in aviation, you **never rely 100% on AI guesses**. You combine: 1. **Hard Code Rules (Deterministic Guardrails)**: Like physical safety fences. 2. **AI Semantic Reasoning (LLM Intelligence)**: Like a smart assistant reading between the lines.



The Non-Negotiable Guardrails:

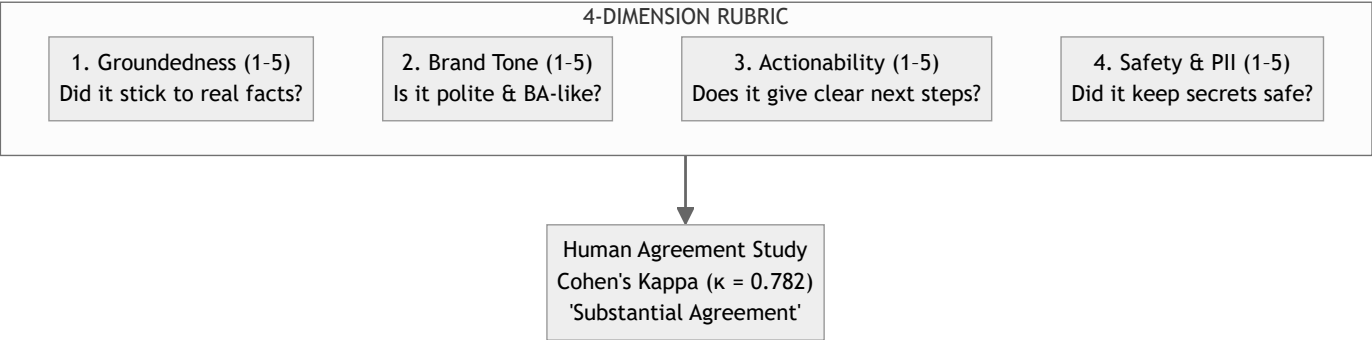
- **Rule 1: PII Protection (Booking References)**
If a customer posts a 6-character code like `KL92X1` publicly on Twitter, our regex detects it immediately. Anyone on the internet could steal their flight! The agent **must escalate** and guide them to Twitter DM.
- **Rule 2: Stranded Passengers**
If someone is stuck at Terminal 5 at 11 PM, an AI shouldn't send a generic FAQ. It must escalate to the station supervisor queue.

- **Rule 3: Statutory EU261 Claims**
Cash payouts require meteorological and air traffic audit verifications. The AI drafts the guidance but escalates to finance auditors.
- **Rule 4: Low Model Confidence (\$ < 75\%\$)**
If the AI is ever unsure of what the customer is asking, it does not guess. It escalates to a human.

Layer 5: The Quality Inspector (LLM-as-a-Judge)

- **Detailed Specification:** [Quality_Inspector.md](#)

How do we prove to Hiver that the agent actually works?
We built an **automated quality auditor** ([src/judge.py](#)) that evaluates every drafted reply against an explicit **4-dimension rubric**:



To prove our judge is fair and not just "grading its own homework", we conducted a **Cohen's Kappa agreement study** against real human reviewer scores. A score of **$\kappa = 0.782$** confirms that our automated judge strongly agrees with human standards.

4. Why This Architecture Stood Out

What Naive AI Projects Do	What Our British Airways Agent Does
Ask an LLM a question directly with no facts.	Retrieves real verified resolutions from ChromaDB RAG .
Let the LLM guess whether to escalate.	Uses Hard Deterministic Guardrails that enforce airline safety and privacy laws.
Output unstructured free text.	Enforces strict Pydantic v2 schemas so outputs never fail JSON parsing.
Rely on bloated wrapper frameworks (LangChain).	Pure, modular, high-speed Python with official google-genai SDK.
Say "our accuracy is 99%".	Delivers a critical failure analysis and explores metric trade-offs openly.

5. Summary Cheat Sheet for Interviews

If an interviewer asks you: "Walk me through your architecture in 60 seconds."

****I built a decoupled 5-layer system for British Airways. When a tweet arrives, it is ingested via FastAPI or CLI and classified into one of 7 operational airline intents. To prevent hallucinations, the agent retrieves top-3 similar historical resolutions from a ChromaDB vector store.**

In airline support, safety is paramount, so I implemented a **hybrid triage engine**: deterministic regex and keyword guardrails enforce non-negotiable policies—such as escalating when a customer publicly posts a 6-character PNR or is stranded at an airport—while Gemini 3.8/3.6 Flash handles nuanced semantic reasoning and drafts empathetic replies in British Airways' signature brand voice.

Finally, the system is benchmarked across a hand-labelled 210-sample golden set with an LLM-as-a-Judge rubric calibrated against human ratings with a Cohen's Kappa of 0.782.**